DRUM COUPLING
ABC-V
B06 20 246 E-EN

B.) Main Hoist

- Max. payload : m_1 = 20000 kg
- Dead weight of the suspension elements : m_2 = 7000 kg
- Dead weight of the rope drum : m_{Tr} = 3000 kg
- Installed motor power : P_i = 450 kW
- Rated motor speed : n_M = 900 min-1
- Gear ratio : i_G = 20
- Rope drum diameter : D_{Tr} = 1.4 m
- Lifting speed : v_H = 90 m/min
- Ratio of reeving : i_F = 2 (see drawing 6)
- Efficiency of reeving : η_F = 0.97
- Engine group : FEM 1.001 = M7
- Operating coefficient : C_{erf} = 1.8

Rotation speed of the rope drum

$$n_{Tr} = \frac{n_M}{i_G} = \frac{900\ \text{min}^{-1}}{20} \qquad \underline{\underline{= 45\ \text{min}^{-1}}}$$

Max. drive torque based on installed power

$$T_{A\,max} = \frac{P_i \cdot 9550}{n_{Tr}} \cdot C_{erf} = \frac{450 \cdot 9550}{45} \cdot 1{,}8 \qquad \underline{\underline{= 171900\ \text{Nm}}}$$

Max. drive torque based on used power

$$T'_{A\,max} = \frac{P_e \cdot 9550}{n_{Tr}} \cdot C_{erf}$$

$$P_e = \frac{S_{Tr} \cdot V_{Tr}}{60000}$$

$$S_{Tr} = \frac{(m_1 + m_2) \cdot 9{,}81}{i_F \cdot \eta_F} = \frac{(20000 + 7000) \cdot 9.81}{2 \cdot 0{,}97} \qquad \underline{\underline{= 136500\ \text{N}}}$$

$$v_{Tr} = v_H \cdot i_F = 90\ \frac{\text{m}}{\text{min}} \cdot 2 \qquad \underline{\underline{= 180\ \text{m/min}}}$$

$$P_e = \frac{136500 \cdot 180}{60000} \qquad \underline{\underline{= 410\ \text{kW}}}$$

$$T'_{A\,max} = \frac{410 \cdot 9550}{45} \cdot 1{,}8 \qquad \underline{\underline{= 156600\ \text{Nm}}}$$

Chosen drum coupling

ABC-V-450 T_{kmax} = 180000 Nm
F_{rmax} = 150000 N

Max. radial load

$$F_{R\,max} = \frac{S_{Tr}}{2} + \frac{m_{Tr} \cdot 9{,}81}{2} = \frac{136500}{2} + \frac{3000 \cdot 9.81}{2} \qquad \underline{\underline{= 83000\ \text{N}}}$$

$\boxed{T'_{Amax} = 156600\ \text{Nm} \le T_{kmax}\ 180000\ \text{Nm}}$ $\boxed{T'_{Amax} = 156600\ \text{Nm} \le T_{kmax}\ 180000\ \text{Nm}}$

Dimension sheet ABC-V

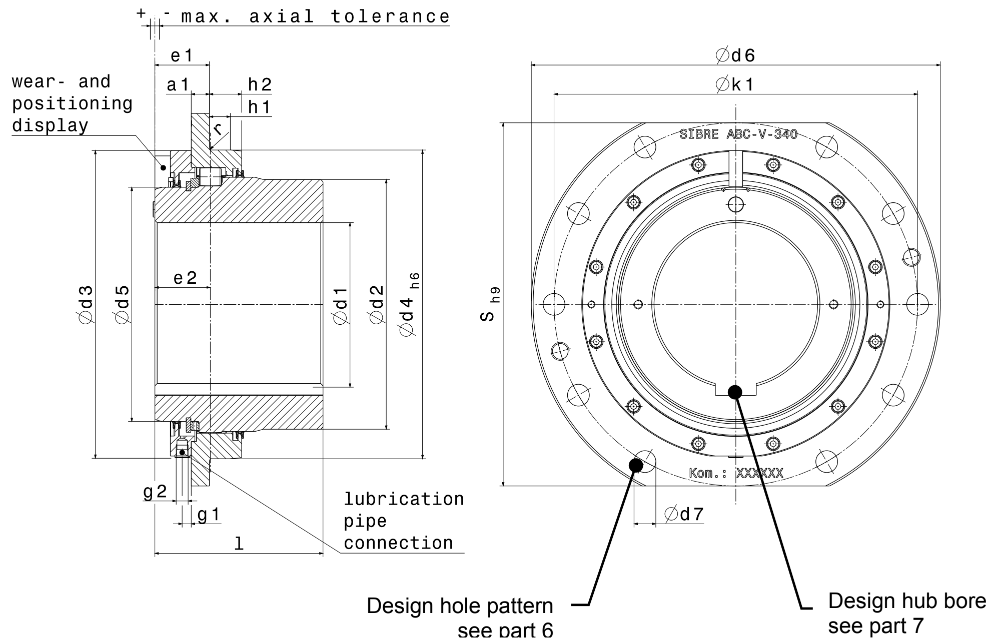

Table 3

Size			260	280	310	340	400	420	450	530	545	560	600	670	730	800	860
Torque[1] $T_{k\,max}$		[Nm]	27000	35000	45000	55000	80000	120000	180000	250000	320000	410000	500000	600000	770000	950000	1025000
Radial load $F_{r\,max}$		[N]	41000	45000	55000	75000	115000	130000	150000	200000	260000	315000	340000	400000	475000	525000	550000
Weight[3]		[kg]	37	44	54	71	108	135	164	260	294	329	415	549	697	960	1097
Moment of inertia[3]		[kgm²]	0,43	0,54	0,82	1,35	2,67	3,7	5,2	11,0	13,2	15,6	22,3	36,3	56,2	105,5	118,4
Finish bore[2]	Ø $d1_{min}^{H7}$	[mm]	80	100	100	100	120	120	140	160	160	170	200	230	260	290	330
	Ø $d1_{max}^{H7}$	[mm]	125	140	155	180	210	215	245	290	300	310	330	370	420	450	470
Ø d2		[mm]	195	215	235	275	315	330	370	430	450	465	500	560	620	680	715
Ø d3		[mm]	259	279	309	339	399	419	449	529	544	558	598	668	728	798	835
Ø $d4_{h6}$		[mm]	260	280	310	340	400	420	450	530	545	560	600	670	730	800	860
Ø d5		[mm]	180	198	218	258	298	310	350	410	430	440	470	530	590	650	680
Ød6		[mm]	380	400	420	450	510	550	580	650	665	680	710	780	850	940	1025
Ø d7		[mm]	19	19	19	24	24	24	24	24	24	24	28	28	28	28	34
a1		[mm]	15	15	15	20	20	20	20	25	25	25	35	35	35	40	40
e1		[mm]	45	45	45	60	60	60	60	65	65	65	81	81	81	86	86
e2		[mm]	48	48	50	61	61	65	67	69	78	78	88	88	90	92	92
g1		[mm]	7.5	7.5	7.5	10	10	10	10	10	10	10	10	10	10	10	10
g2		[In]	G1/8	G1/8	G1/8	G1/4	G1/4	G1/4	G1/4	G1/4	G1/4	G1/4	G1/4	G1/4	G1/4	G1/4	G1/4
h1		[mm]	27,5	27,5	27	23	24,5	30	32	35	45	45	40	40	50	50	50
h2		[mm]	34,5	34,5	37	35,5	37	45	47	50	60	65	60	60	70	70	70
S_{h9}		[mm]	340	360	380	400	460	500	530	580	590	600	640	700	760	830	900
Ø k1		[mm]	340	360	380	400	460	500	530	600	615	630	660	730	800	875	945
l		[mm]	145	170	175	185	220	240	260	315	330	350	380	410	450	500	500
r		[mm]	2	2.5	2.5	2.5	2.5	2.5	2.5	2.5	4	4	4	4	4	4	4
Axial tolerance max +/-		[mm]	4	4	4	5	6	6	6	6	6	6	8	8	8	10	10

(1) The given torques do not refer to the shaft-hub connection. These must be checked if necessary.
(2) Other tolerances possible by arrangement.
(3) With respect to max. finish bore Ød1.

Alterations reserved – SIBRE Siegerland-Bremsen GmbH – Auf der Stücke 1-5 – D-35708 Haiger, Germany
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